

(A) Experiment selection

Choose an experiment

Analysis

Full benchmark

Dataset / target

biodolphin Kd

Representation

ChemBERTa-MLM - 384D

Kernel

Matérn 3/2

Acquisition policy

Joint Thompson sampling

Seed

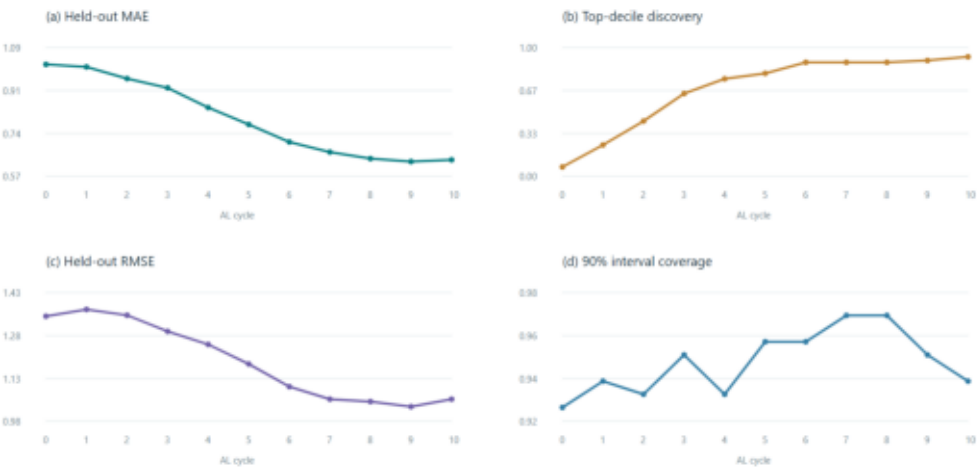
7

5850 completed benchmark runs - 90 explanation reports

(C) Matched representation comparison

Representation	Seeds	MAE $\pm$ SD	R ² $\pm$ SD	Top-10% recall $\pm$ SD	Recall AUC $\pm$ SD
ECFP - radius 4 / 4096 bits	3	0.759 $\pm$ 0.133	0.578 $\pm$ 0.103	0.924 $\pm$ 0.050	0.677 $\pm$ 0.097
MACCS - 167 bits	3	0.753 $\pm$ 0.108	0.565 $\pm$ 0.095	0.852 $\pm$ 0.139	0.660 $\pm$ 0.078
ChemBERTa-MTR - 384D	3	0.705 $\pm$ 0.046	0.624 $\pm$ 0.007	0.967 $\pm$ 0.016	0.711 $\pm$ 0.033
ChemBERTa-MLM - 384D	3	0.711 $\pm$ 0.074	0.620 $\pm$ 0.035	0.919 $\pm$ 0.030	0.670 $\pm$ 0.029
MolFormer-XL - 768D	3	0.753 $\pm$ 0.093	0.561 $\pm$ 0.060	0.843 $\pm$ 0.080	0.642 $\pm$ 0.055

(B) Four-panel learning curves



(D) Local upload campaign controls

1. Upload measured pairs

CSV file - 5 MB maximum - 10-1000 rows

Choose File No file chosen

Required columns: protein_sequence, lipid_smiles, affinity.

Use one affinity endpoint, standard amino-acid sequences and valid SMILES. Every affinity must be measured and numeric. Duplicates are rejected after canonicalizing SMILES.

2. Configure the experiment

Affinity units

pKd / pKi (larger is better)

Lipid representation

MolFormer-XL - 768D

Lipid kernel

tanimoto

Acquisition policy

Thompson + diversity heuristic

Initial measured pairs

10

Pairs acquired per cycle

5

Acquisition cycles

5

Random seed

7

Transformers run locally using downloaded frozen weights. EI/VI use latent uncertainty and $\xi_i = 0.01$ pK (marginal top-k batch, not qEI). Thompson + diversity uses a fixed 0.25 diversity bonus. All five lipid kernels are available for all five representations.

A frozen 20% random-pair test set is excluded from acquisition. Initial + batch $\times$ cycles must fit within the remaining pool (maximum 500 acquired labels). Raw Kd/Ki is converted to pK. Protein composition uses a Matérn 3/2 factor.

This is retrospective AL, not a live experimental oracle. Input data remain on this computer under results/user_runs. New runs produce metrics, query indices and test predictions; SHAP is not automatically computed for uploads.

Validate & start campaign -- No benchmark results are overwritten.